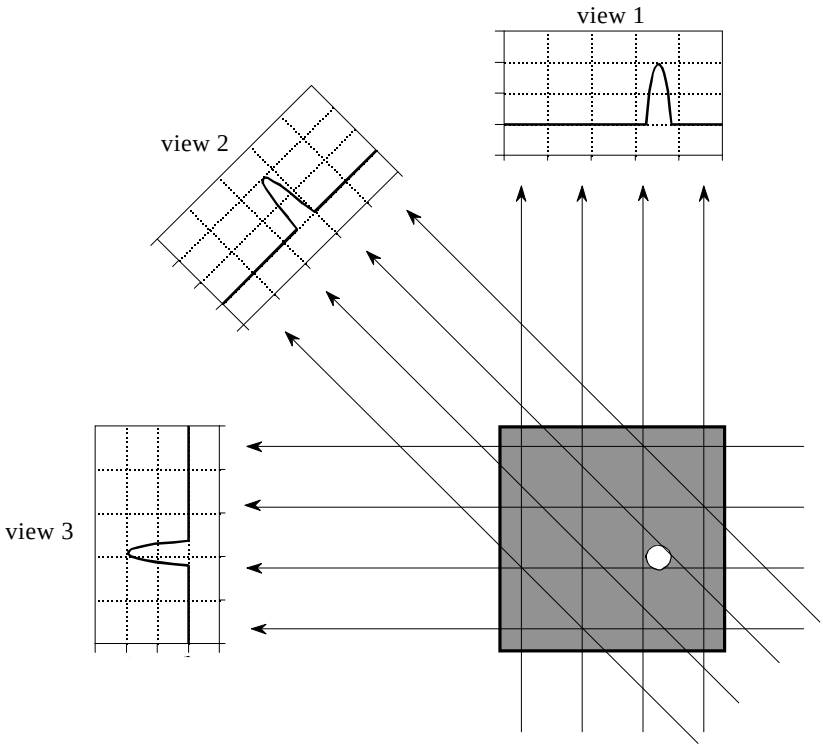

FIGURE 25-15
CT views. Computed tomography acquires a set of views and then reconstructs the corresponding image. Each sample in a view is equal to the sum of the image values along the ray that points to that sample. In this example, the image is a small pillbox surrounded by zeros. While only three views are shown here, a typical CT scan uses hundreds of views at slightly different angles.

sum of the image pixels along the ray pointing to the sample. If the ray sum is lower than the measured sample, all the pixels along the ray are increased in value. Likewise, if the ray sum is higher than the measured sample, all of the pixel values along the ray are decreased. After the first complete iteration cycle, there will still be an error between the ray sums and the measured values. This is because the changes made for any one measurement disrupts all the previous corrections made. The idea is that the errors become smaller with repeated iterations until the image converges to the proper solution.

Iterative techniques are generally slow, but they are useful when better algorithms are not available. In fact, ART was used in the first commercial medical CT scanner released in 1972, the EMI Mark I. We will revisit iterative techniques in the next chapter on neural networks. Development of the third and forth methods have almost entirely replaced iterative techniques in commercial CT products.

The last two reconstruction algorithms are based on formal mathematical solutions to the problem. These are elegant examples of DSP. The third method is called **filtered backprojection**. It is a modification of an older